

Algoritmo		Handshake		Pacotes Trocados [ms]	
Troca de Chave	Assinatura	Latência [ms]	Amplificação Realizada	Número	Tamanho
BP384 + ML-KEM-768	P384 + ML-DSA-65	10.1±0.4	2.91x	6 / 12	3778 / 1
BP384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	20.6±0.3	18.01x	17 / 66	4229 / 7
BP384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	221.5±1.1	9.17x	9 / 32	3901 / 3
BP384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	31.6±0.4	18.19x	16 / 66	4188 / 7
BP384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	463.4±1.0	9.17x	9 / 32	3901 / 3
P256 + ML-KEM-768	P384 + ML-DSA-65	4.7±0.5	2.89x	6 / 13	3808 / 1
P256 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	15.1±0.5	19.11x	11 / 66	3983 / 7
P256 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	215.7±1.3	9.36x	7 / 32	3819 / 3
P256 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	26.0±0.6	18.73x	13 / 66	4065 / 7
P256 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	456.6±0.9	9.36x	7 / 32	3819 / 3
P384 + ML-KEM-768	P384 + ML-DSA-65	9.8±0.3	2.91x	6 / 12	3778 / 1
P384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	20.3±0.1	18.37x	15 / 66	4147 / 7
P384 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	220.3±0.3	9.17x	9 / 32	3901 / 3
P384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	31.0±0.1	18.93x	12 / 66	4024 / 7
P384 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	460.5±0.2	9.27x	8 / 32	3860 / 3
P384 + BIKE-L3	P384 + ML-DSA-65	12.0±0.4	2.65x	6 / 13	4937 / 1
P384 + BIKE-L3	P384 + SPHINCS-SHA2-192-F-SIMPLE	22.4±0.5	14.75x	15 / 68	5306 / 7
P384 + BIKE-L3	P384 + SPHINCS-SHA2-192-S-SIMPLE	222.1±0.2	7.55x	8 / 34	5019 / 3
P384 + BIKE-L3	P384 + SPHINCS-SHAKE-192-F-SIMPLE	33.7±0.8	15.10x	12 / 68	5183 / 7
P384 + BIKE-L3	P384 + SPHINCS-SHAKE-192-S-SIMPLE	463.2±0.7	7.55x	8 / 34	5019 / 3
P384 + FRODO-976-AES	P384 + ML-DSA-65	14.9±2.4	1.25x	21 / 27	22967 / 1
P384 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-F-SIMPLE	31.5±6.8	3.46x	29 / 80	26742 / 1
P384 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-S-SIMPLE	222.8±1.3	1.88x	28 / 46	27860 / 1
P384 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-F-SIMPLE	34.5±1.3	3.32x	29 / 81	27933 / 1
P384 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-S-SIMPLE	464.0±2.1	1.80x	29 / 46	29060 / 1
P384 + FRODO-976-SHAKE	P384 + ML-DSA-65	20.2±3.5	1.24x	22 / 28	24111 / 1
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-F-SIMPLE	28.7±5.5	3.45x	31 / 81	26856 / 1
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-S-SIMPLE	227.3±2.4	1.88x	27 / 46	27820 / 1
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-F-SIMPLE	37.0±1.4	3.46x	30 / 80	26783 / 1
P384 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-S-SIMPLE	467.0±1.9	1.80x	29 / 46	29060 / 1
P384 + HQC-192	P384 + ML-DSA-65	81.6±3.9	2.59x	9 / 18	7378 / 1
P384 + HQC-192	P384 + SPHINCS-SHA2-192-F-SIMPLE	91.2±0.6	11.06x	15 / 73	7624 / 8
P384 + HQC-192	P384 + SPHINCS-SHA2-192-S-SIMPLE	295.1±2.8	5.90x	11 / 39	7460 / 4
P384 + HQC-192	P384 + SPHINCS-SHAKE-192-F-SIMPLE	117.0±16.5	9.52x	17 / 74	8865 / 8
P384 + HQC-192	P384 + SPHINCS-SHAKE-192-S-SIMPLE	541.9±4.5	5.90x	11 / 39	7460 / 4
X25519 + ML-KEM-768	P384 + ML-DSA-65	5.0±0.5	2.88x	6 / 13	3808 / 1
X25519 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	15.4±0.7	19.14x	11 / 67	3978 / 7
X25519 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	215.1±0.6	9.29x	7 / 33	3849 / 3
X25519 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	25.6±0.3	18.53x	14 / 66	4106 / 7
X25519 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	455.7±0.3	9.36x	7 / 33	3820 / 3
X448 + ML-KEM-768	P384 + ML-DSA-65	6.0±0.7	2.89x	6 / 13	3808 / 1
X448 + ML-KEM-768	P384 + SPHINCS-SHA2-192-F-SIMPLE	15.7±0.3	19.11x	11 / 66	3983 / 7
X448 + ML-KEM-768	P384 + SPHINCS-SHA2-192-S-SIMPLE	216.2±0.7	9.36x	7 / 32	3819 / 3
X448 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-F-SIMPLE	26.7±0.8	19.11x	11 / 66	3983 / 7
X448 + ML-KEM-768	P384 + SPHINCS-SHAKE-192-S-SIMPLE	456.6±0.3	9.26x	8 / 32	3860 / 3
X448 + BIKE-L3	P384 + ML-DSA-65	7.2±0.2	2.62x	7 / 13	4978 / 1
X448 + BIKE-L3	P384 + SPHINCS-SHA2-192-F-SIMPLE	18.1±0.5	15.09x	12 / 68	5183 / 7
X448 + BIKE-L3	P384 + SPHINCS-SHA2-192-S-SIMPLE	217.8±0.6	7.54x	8 / 34	5019 / 3
X448 + BIKE-L3	P384 + SPHINCS-SHAKE-192-F-SIMPLE	29.9±0.7	14.98x	13 / 68	5224 / 7
X448 + BIKE-L3	P384 + SPHINCS-SHAKE-192-S-SIMPLE	459.5±1.1	7.51x	8 / 35	5049 / 3
X448 + FRODO-976-AES	P384 + ML-DSA-65	7.8±0.6	1.12x	24 / 27	25408 / 1
X448 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-F-SIMPLE	27.3±0.8	3.47x	27 / 81	26692 / 1
X448 + FRODO-976-AES	P384 + SPHINCS-SHA2-192-S-SIMPLE	218.8±2.0	1.87x	28 / 46	27860 / 1
X448 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-F-SIMPLE	28.2±0.4	3.46x	29 / 81	26774 / 1
X448 + FRODO-976-AES	P384 + SPHINCS-SHAKE-192-S-SIMPLE	459.0±1.4	1.80x	29 / 46	29060 / 1
X448 + FRODO-976-SHAKE	P384 + ML-DSA-65	13.2±0.5	1.18x	22 / 27	24167 / 1
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-F-SIMPLE	28.6±6.4	3.46x	29 / 81	26774 / 1
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHA2-192-S-SIMPLE	221.2±1.3	1.87x	29 / 46	27886 / 1
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-F-SIMPLE	34.1±1.0	3.63x	27 / 80	25501 / 1
X448 + FRODO-976-SHAKE	P384 + SPHINCS-SHAKE-192-S-SIMPLE	463.5±1.4	1.80x	27 / 46	28979 / 1
X448 + HQC-192	P384 + ML-DSA-65	74.2±0.4	2.59x	9 / 18	7378 / 1
X448 + HQC-192	P384 + SPHINCS-SHA2-192-F-SIMPLE	86.7±1.6	10.77x	20 / 73	7829 / 8
X448 + HQC-192	P384 + SPHINCS-SHA2-192-S-SIMPLE	293.0±10.2	5.89x	11 / 39	7460 / 4
X448 + HQC-192	P384 + SPHINCS-SHAKE-192-F-SIMPLE	97.4±1.7	10.04x	17 / 73	7706 / 4